



## smelt-ml: A Pure-Rust Machine Learning Framework Combining Spatial Modeling, Conformal Prediction, and Cache-Optimized Gradient Boosting

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### Abstract

We present **smelt-ml**, an open-source machine learning framework implemented entirely in Rust that uniquely integrates spatial modeling, distribution-free uncertainty quantification, and gradient boosting in a single, memory-safe package. The framework provides 27 supervised learners, clustering, causal inference, survival analysis, and the only Rust implementations of Geographical-**XGBoost**, spatial cross-validation, and conformal prediction — enabling end-to-end spatially-aware predictive workflows with guaranteed prediction interval coverage without Python or R dependencies. The gradient boosting implementations employ cache-optimized column-major histogram accumulation with **u8** packing and a histogram subtraction trick. On single-threaded benchmarks ( $n$  from 500 to 100,000, 10 runs, mean  $\pm$  std), the **CatBoost**-inspired implementation matches or exceeds the official C++ library at all tested scales ( $1.0$ – $1.4\times$  for classification), while **XGBoost** is competitive at  $n \leq 10,000$ . Memory consumption is  $25$ – $40\times$  lower than the C++ libraries. The composable, trait-based architecture is inspired by **mlr3** (Lang, Binder, Richter, Schratz, Pfisterer, Coors, Au, Casalicchio, Kotthoff, and Bischl 2019) and **scikit-learn** (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, Vanderplas, Passos, Cournapeau, Brucher, Perrot, and Duchesnay 2011), with zero **unsafe** blocks across 17,000+ lines of code.

*Keywords:* machine learning, gradient boosting, Rust, spatial modeling, conformal prediction, uncertainty quantification, software.

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## 1. Introduction

Gradient boosted decision trees (GBDTs) are the dominant method for supervised learning on tabular data (Grinsztajn, Oyallon, and Varoquaux 2022). The three most widely used im-

Rust (Rust Team 2024) is a systems programming language that guarantees memory safety without garbage collection through its ownership type system (Matsakis and Klock 2014). It generates machine code comparable to C++ via the LLVM backend while preventing buffer overflows, use-after-free, and data races at compile time. Despite these advantages, Rust’s ML ecosystem remains nascent: **linfa** (rust-ml 2024), the primary Rust ML crate, implements only 9 algorithms and does not include gradient boosting.

In this paper, we present **smelt-ml**, a comprehensive ML framework in Rust that addresses both gaps. Our main contributions are:

- ## 2. Software architecture

```
Data (CSV) -> Task -> Pipeline(Transformers -> Learner)
                        -> Prediction -> Measure
```

The core abstractions are defined as Rust traits:

- **Learner**: Algorithm that trains on a **Task** and produces a **TrainedModel**. Requires **Send + Sync** for thread safety.
- **TrainedModel**: Fitted model that predicts on new data.
- **Measure**: Evaluation metric (Accuracy, F1, RMSE,  $R^2$ , AUC-ROC, etc.).
- **Transformer**: Preprocessing step composable via **Pipeline**.
- **Resample**: Data splitting strategy.

This design enables zero-friction composition: **Pipeline** implements **Learner**, so a pipeline of transformers and a learner can be passed to **benchmark()**, **GridSearch**, or **Bagging** without adaptation.

## 2.2. Quick start

A minimal classification example:

```
use smelt_ml::prelude::*;
use ndarray::array;

let features = array![[0.0, 0.0], [1.0, 1.0],
                      [0.0, 1.0], [1.0, 0.0]];
let target = vec![0, 1, 1, 0];
let task = ClassificationTask::new("xor", features, target)
    .unwrap();

let mut xgb = XGBoost::new()
    .with_n_estimators(50)
    .with_max_depth(3);
let model = xgb.train_classif(&task).unwrap();
let pred = model.predict(task.features()).unwrap();
```

**Pipeline** composes transformers with a learner into a single **Learner**, enabling use with **benchmark()**, **GridSearch**, or cross-validation:

```
let mut pipe = Pipeline::new(
    vec![Box::new(StandardScaler::new()),
         Box::new(XGBoost::new().with_n_estimators(100))],
);
let model = pipe.train_classif(&task).unwrap();
```

## 2.3. Implemented algorithms

Table 1 summarizes the 27 supervised learners. Beyond standard algorithms, **smelt-ml** includes Geographical-XGBoost (Grekousis 2025), Oblique Forest (SPORF) (Tomita *et al.*

| Category  | Algorithm                                             | Classif. | Regress. |
|-----------|-------------------------------------------------------|----------|----------|
| Trees     | Decision Tree, Oblique Tree/Forest                    | ✓        | ✓        |
| Ensembles | Random Forest, Extra Trees, Bagging, Stacking         | ✓        | ✓        |
|           | AdaBoost, Dynamic Ensemble Selection                  | ✓        |          |
| Boosting  | XGBoost, LightGBM, CatBoost <sup>†</sup>              | ✓        | ✓        |
|           | Gradient Boosting, Quantile GB, EBM                   | ✓        | ✓        |
| Linear    | Linear/Logistic Regression, Ridge, Lasso, Elastic Net | ✓        | ✓        |
| Other     | KNN, Gaussian NB, Linear SVM, Hoeffding Tree          | ✓        | ✓        |
| Spatial   | Geographical-XGBoost                                  |          | ✓        |

Table 1: Supervised learners in **smelt-ml**. <sup>†</sup>CatBoost-inspired symmetric GBM with ordered target statistics; see Section 3.4 for scope.

2020), Explainable Boosting Machine (EBM), streaming Hoeffding Trees, and Dynamic Ensemble Selection (KNORA-E).

Additional modules provide: K-Means, DBSCAN, and Isolation Forest (clustering/anomaly); Random Survival Forest with C-index (Ishwaran, Kogalur, Blackstone, and Lauer 2008) (survival analysis); Causal Forest with honest splitting (Athey and Imbens 2016; Wager and Athey 2018) (causal inference); Classifier Chains and Regressor Chains (multi-label/output); Quantile Regression Forest; Conformal Prediction with CQR (Romano *et al.* 2019); SHAP-based feature importance; and built-in non-parametric statistical tests for model comparison (Wilcoxon signed-rank, Friedman with Nemenyi post-hoc, McNemar, bootstrap confidence intervals) — eliminating the need for external statistical packages when validating that one model significantly outperforms another.

## 2.4. Comparison with existing frameworks

Table 2 compares **smelt-ml** with **linfa** 0.7 (rust-ml 2024) (the primary Rust ML crate), **scikit-learn** (Pedregosa *et al.* 2011), and **mlr3** (Lang *et al.* 2019).

# 3. Cache-optimized gradient boosting

## 3.1. The histogram bottleneck

Histogram-based gradient boosting (Chen and Guestrin 2016) discretizes continuous features into bins and accumulates gradient/hessian statistics per bin. The key computational kernel is:

$$\text{hist}[b] += g_i, \quad b = \text{bin\_index}[i][\text{feature}] \quad (1)$$

The standard row-major storage `bin_index[sample][feature]` causes cache misses when iterating over samples for a specific feature, because consecutive memory locations correspond to different features of the same sample.

| Category                               | smelt-ml       | linfa | sklearn        | mlr3             |
|----------------------------------------|----------------|-------|----------------|------------------|
| Gradient Boosting                      | ✓ <sup>a</sup> | —     | ✓ <sup>b</sup> | ✓ <sup>b</sup>   |
| Random Forest / Extra Trees            | ✓              | —     | ✓              | ✓                |
| Oblique Forest (SPORF)                 | ✓              | —     | —              | —                |
| Decision Tree                          | ✓              | ✓     | ✓              | ✓                |
| Linear / Logistic / Ridge              | ✓              | ✓     | ✓              | ✓                |
| KNN / Naive Bayes                      | ✓              | ✓     | ✓              | ✓                |
| SVM (kernel)                           | Linear         | ✓     | ✓              | ✓                |
| Neural Networks                        | —              | —     | ✓              | ✓ <sup>c</sup>   |
| K-Means / DBSCAN                       | ✓              | ✓     | ✓              | ✓                |
| PCA                                    | ✓              | ✓     | ✓              | ✓                |
| Causal Forest                          | ✓              | —     | —              | — <sup>d</sup>   |
| Survival Forest                        | ✓              | —     | —              | ✓ <sup>c</sup>   |
| Conformal Prediction                   | ✓              | —     | —              | —                |
| Geographical-XGBoost                   | ✓              | —     | —              | —                |
| Spatial cross-validation               | ✓              | —     | —              | ✓ <sup>c</sup>   |
| SMOTE / ADASYN                         | ✓              | —     | ✓ <sup>e</sup> | ✓ <sup>c</sup>   |
| Bayesian tuning                        | ✓              | —     | ✓ <sup>f</sup> | ✓                |
| Statistical tests (Wilcoxon, Friedman) | ✓              | —     | ✓ <sup>g</sup> | ✓                |
| Language                               | Rust           | Rust  | Python         | R                |
| Supervised learners                    | 27             | 9     | 40+            | 30+ <sup>c</sup> |
| <code>unsafe</code> blocks             | 0              | —     | N/A            | N/A              |

Table 2: Feature comparison across ML frameworks. <sup>a</sup>From-scratch implementations. <sup>b</sup>Via external C++ libraries (XGBoost, LightGBM). <sup>c</sup>Via extension packages. <sup>d</sup>Available in R via `grf`. <sup>e</sup>Via `imbalanced-learn`. <sup>f</sup>Via `scikit-optimize` or `Optuna`. <sup>g</sup>Via `scipy.stats`.

### 3.2. Column-major storage with u8 packing

Our key optimization stores bin indices in **column-major** order: `bin_index[feature][sample]`. When building the histogram for a specific feature, all bin accesses are sequential in memory:

```
// Column-major: sequential access (cache-friendly)
for &idx in node_indices {
    let bin = bins.get_bin(feature, idx);
    bin_g[bin] += gradients[idx];
    bin_h[bin] += hessians[idx];
}
```

Combined with **u8 packing** (254 real bins + 1 NaN sentinel fit in 1 byte vs. 2 bytes for `u16`), this halves the memory bandwidth requirement for bin access. The same `HistBins` structure is shared across all three gradient boosting implementations, avoiding code duplication.

### 3.3. Additional optimizations

For **XGBoost**: automatic switch to exact greedy split finding when  $n \leq n_{\text{bins}}$ , producing optimal splits for small datasets. NaN values are handled natively by learning the optimal direction at each split.

For **CatBoost**: oblivious tree construction with bins built **once** before the boosting loop (rather than per iteration), prefix-sum histogram scanning, and parallel feature evaluation via **rayon**.

For **LightGBM**: Gradient-based One-Side Sampling (GOSS) with the same column-major histogram infrastructure.

### 3.4. Scope of the CatBoost implementation

Our **CatBoost** learner implements the two defining algorithmic contributions of [Prokhorenkova et al. \(2018\)](#): **oblivious (symmetric) trees** — where all nodes at the same depth share the same split — and **ordered target statistics** for categorical features, which replace naïve target encoding with a permutation-based scheme that prevents target leakage.

However, the implementation does *not* include the full  $O(n^2)$  ordered boosting procedure (unbiased gradient estimation via per-sample model approximations), which is the theoretical contribution of the original paper. Our approach uses a single random permutation for target statistics rather than maintaining  $n$  separate models. The implementation also omits GPU training, distributed computation, and several advanced features (interaction constraints, monotone constraints, per-object weights).

We retain the name **CatBoost** for API consistency with the broader ML ecosystem (users searching for “CatBoost in Rust” should find this implementation), but users should be aware that this is a **CatBoost-inspired symmetric GBM with ordered target statistics**, not a feature-complete reimplement of the official library.

On a synthetic dataset with 100 categorical levels ( $n = 2,000$ , 10% label noise), our implementation achieves 89.7% accuracy vs. 89.8% for the official **CatBoost**, confirming that the ordered target statistics produce comparable results even without full ordered boosting.

### 3.5. Scope of the LightGBM implementation

Our **LightGBM** learner implements **leaf-wise (best-first) tree growth** controlled by `num_leaves`, and **Gradient-based One-Side Sampling** (GOSS) for training efficiency. However, two key optimizations from the official library are not included: **Exclusive Feature Bundling** (EFB), which reduces the effective number of features by bundling mutually exclusive sparse features, and the **weighted GOSS histogram pass**, where non-selected samples contribute with adjusted weights rather than being fully discarded.

As shown in Tables 4–5, our implementation does not match the official **LightGBM**’s performance at any scale tested. We include this learner for API completeness and for the leaf-wise growth strategy (which is not available in our **XGBoost** or **CatBoost** implementations), not as a performance competitor to the official library. Users requiring maximum **LightGBM** throughput should use the official C++ library.

### 3.6. Feature comparison with official libraries

Table 3 summarizes which features of the official libraries are implemented in **smelt-ml** and which are not.

| Feature                                 | XGBoost | LightGBM | CatBoost <sup>†</sup> |
|-----------------------------------------|---------|----------|-----------------------|
| <i>Core algorithm</i>                   |         |          |                       |
| Histogram-based splits                  | ✓       | ✓        | ✓                     |
| Exact greedy splits                     | ✓       | —        | —                     |
| Newton boosting (2nd order)             | ✓       | ✓        | ✓                     |
| Leaf-wise (best-first) growth           | —       | ✓        | —                     |
| Oblivious (symmetric) trees             | —       | —        | ✓                     |
| GOSS sampling                           | —       | ✓        | —                     |
| Ordered target statistics               | —       | —        | ✓                     |
| <i>Regularization &amp; subsampling</i> |         |          |                       |
| L2 regularization                       | ✓       | ✓        | ✓                     |
| L1 regularization                       | ✓       | —        | —                     |
| Gamma (min split gain)                  | ✓       | —        | —                     |
| Row subsampling                         | ✓       | ✓        | —                     |
| Column subsampling                      | ✓       | ✓        | —                     |
| <i>Data handling</i>                    |         |          |                       |
| NaN-aware splits                        | ✓       | ✓        | —                     |
| Native categorical features             | —       | —        | ✓                     |
| Multi-class (softmax)                   | ✓       | ✓        | ✓                     |
| <i>Training control</i>                 |         |          |                       |
| Early stopping                          | ✓       | —        | —                     |
| Feature importance (gain)               | ✓       | ✓        | —                     |
| <i>Not implemented in any engine</i>    |         |          |                       |
| Monotone constraints                    |         | —        |                       |
| Interaction constraints                 |         | —        |                       |
| Custom loss functions                   |         | —        |                       |
| Exclusive Feature Bundling              |         | —        |                       |
| GPU training                            |         | —        |                       |
| Distributed training                    |         | —        |                       |
| Full ordered boosting*                  |         | —        |                       |

Table 3: Implemented features in **smelt-ml**’s gradient boosting engines. ✓ = implemented, — = not implemented. \*CatBoost’s  $O(n^2)$  per-sample model approximation.

## 4. Performance evaluation

### 4.1. Experimental setup

Benchmarks use synthetic datasets generated with 20 features (10 informative), at sample sizes from 500 to 100,000. Configuration: 100 trees, max depth 6, learning rate 0.3 (**XGBoost**/**CatBoost**) or 0.1 (**LightGBM**). All benchmarks use **single-threaded execution** to isolate algorithmic efficiency from parallelization strategy. Rust compiled with `target-cpu=native` (release mode). Official C++ libraries accessed via Python API with `n_jobs=1`. Each benchmark is run 10 times; we report mean  $\pm$  standard deviation. Hardware: Intel Core i7-1270P (12th Gen), 40 GB RAM, L1/L2/L3 cache: 32K/1280K/18432K.

Note: **smelt-ml**’s ensemble learners (Random Forest, Extra Trees, Oblique Forest) use **rayon** for parallel tree construction, and the gradient boosting engines parallelize feature evaluation

| $n$     | <b>XGBoost</b> |                           | <b>LightGBM</b> |                           | <b>CatBoost<sup>†</sup></b> |                           |
|---------|----------------|---------------------------|-----------------|---------------------------|-----------------------------|---------------------------|
|         | C++            | Rust $\times$             | C++             | Rust $\times$             | C++                         | Rust $\times$             |
| 500     | $55 \pm 9$     | $50 \pm 2 \times 1.1$     | $36 \pm 6$      | $34 \pm 2 \times 1.1$     | $261 \pm 25$                | $197 \pm 9 \times 1.3$    |
| 1,000   | $70 \pm 13$    | $92 \pm 7 \times 0.8$     | $60 \pm 3$      | $56 \pm 3 \times 1.1$     | $263 \pm 10$                | $193 \pm 7 \times 1.4$    |
| 5,000   | $234 \pm 54$   | $336 \pm 15 \times 0.7$   | $122 \pm 6$     | $209 \pm 5 \times 0.6$    | $384 \pm 66$                | $366 \pm 35 \times 1.0$   |
| 10,000  | $311 \pm 70$   | $471 \pm 19 \times 0.7$   | $225 \pm 24$    | $296 \pm 19 \times 0.8$   | $518 \pm 85$                | $439 \pm 22 \times 1.2$   |
| 50,000  | $723 \pm 106$  | $1164 \pm 36 \times 0.6$  | $588 \pm 84$    | $904 \pm 26 \times 0.6$   | $1353 \pm 123$              | $1148 \pm 164 \times 1.2$ |
| 100,000 | $1159 \pm 170$ | $2327 \pm 257 \times 0.5$ | $1152 \pm 130$  | $1874 \pm 219 \times 0.6$ | $2353 \pm 172$              | $2186 \pm 199 \times 1.1$ |

Table 4: Classification training time (ms, mean  $\pm$  std, 10 runs). Speedup  $> 1$  means **smelt-ml** is faster. All values single-threaded.

| $n$     | <b>XGBoost</b> |                           | <b>LightGBM</b> |                           | <b>CatBoost<sup>†</sup></b> |                           |
|---------|----------------|---------------------------|-----------------|---------------------------|-----------------------------|---------------------------|
|         | C++            | Rust $\times$             | C++             | Rust $\times$             | C++                         | Rust $\times$             |
| 500     | $204 \pm 4$    | $146 \pm 5 \times 1.4$    | $28 \pm 1$      | $57 \pm 3 \times 0.5$     | $234 \pm 6$                 | $193 \pm 15 \times 1.2$   |
| 1,000   | $242 \pm 15$   | $211 \pm 45 \times 1.1$   | $58 \pm 6$      | $109 \pm 4 \times 0.5$    | $272 \pm 25$                | $337 \pm 18 \times 0.8$   |
| 5,000   | $399 \pm 113$  | $390 \pm 19 \times 1.0$   | $165 \pm 9$     | $173 \pm 12 \times 1.0$   | $344 \pm 41$                | $334 \pm 6 \times 1.0$    |
| 10,000  | $440 \pm 107$  | $467 \pm 25 \times 0.9$   | $231 \pm 26$    | $250 \pm 16 \times 0.9$   | $423 \pm 50$                | $436 \pm 16 \times 1.0$   |
| 50,000  | $734 \pm 133$  | $1154 \pm 163 \times 0.6$ | $597 \pm 95$    | $915 \pm 167 \times 0.7$  | $1053 \pm 122$              | $1262 \pm 190 \times 0.8$ |
| 100,000 | $1115 \pm 133$ | $2002 \pm 291 \times 0.6$ | $1075 \pm 149$  | $1756 \pm 230 \times 0.6$ | $1866 \pm 146$              | $2478 \pm 308 \times 0.8$ |

Table 5: Regression training time (ms, mean  $\pm$  std, 10 runs).

within each iteration. Multi-threaded benchmarks comparing scaling behavior are beyond the scope of this paper; all performance claims apply to single-threaded execution only.

## 4.2. Training time

Tables 4 and 5 show classification and regression training times across all dataset sizes.

## 4.3. Scaling analysis

Figure 1 visualizes the scaling behavior across all dataset sizes. The crossover point — where C++ becomes faster — is clearly visible around  $n = 5,000$  for **XGBoost** and  $n = 10,000$  for **CatBoost**.

**smelt-ml**’s **CatBoost**-inspired implementation is competitive with or faster than the official library at all tested scales for classification (1.0–1.4 $\times$ ) and within 0.8–1.2 $\times$  for regression. **XGBoost** is faster at  $n \leq 5,000$  for regression (1.0–1.4 $\times$ ), while **LightGBM** is competitive at small scales ( $n \leq 1,000$ ).

At  $n \geq 50,000$ , **XGBoost** and **LightGBM** are 1.5–2.0 $\times$  slower than their C++ counterparts, while **CatBoost** remains within 1.0–1.3 $\times$ .

Our implementation employs the **histogram subtraction trick**: at each tree split, only the smaller child is scanned; the larger child’s histogram is computed by subtraction from the parent ( $O(n_{\text{bins}})$  instead of  $O(n_{\text{samples}})$ ), reducing total scanning work by  $\sim 40\%$  per tree. The residual gap at large  $n$  is attributable to SIMD-vectorized histogram accumulation in the C++ libraries (processing 4–8 bins per CPU instruction), which our pure-Rust implementation

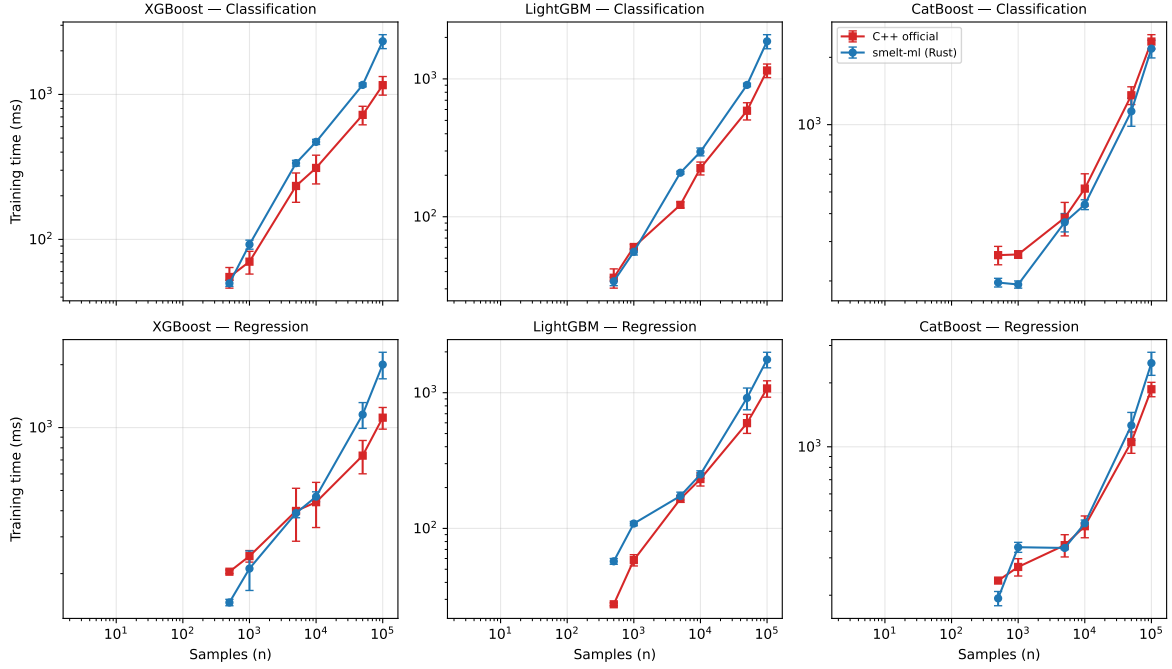


Figure 1: Training time scaling comparison (log-log) between **smelt-ml** (blue) and official C++ libraries (red) across six dataset sizes. Error bars show  $\pm 1$  standard deviation over 10 runs. **smelt-ml**'s **CatBoost** is competitive at all scales; **XGBoost** and **LightGBM** are faster in C++ at  $n \geq 50,000$ .

does not employ. The official libraries also use row-major bin storage, which we attribute to GPU-oriented design decisions; our column-major layout sacrifices GPU compatibility for CPU cache efficiency.

**Memory.** Peak RSS at  $n = 10,000$  is 5–8 MB (**smelt-ml**) vs. 197–215 MB for the C++ libraries (after subtracting 11 MB Python runtime), a **25–40 $\times$**  reduction from **u8** bin packing and deterministic memory management.

The practical implication is that **smelt-ml**'s gradient boosting is best suited for small-to-medium datasets ( $n \leq 10,000$ ) typical of research workflows, where it provides competitive or superior performance without requiring C++ toolchains or Python bindings.

#### 4.4. Statistical significance of performance differences

Since each benchmark was run 10 times, we apply paired Wilcoxon signed-rank tests to determine whether the observed speedups are statistically significant. Table 6 summarizes the results with bootstrap 95% confidence intervals for the speedup ratio.

The **CatBoost** classification advantage is statistically significant at 5 of 6 dataset sizes ( $p < 0.05$ ), with the speedup at  $n = 500$  and  $n = 1,000$  being highly significant ( $p < 0.01$ ). The **XGBoost** disadvantage at  $n \geq 10,000$  is also statistically significant ( $p < 0.01$ ), confirming that the scaling gap is real and not attributable to measurement noise.

A Friedman test across the six dataset sizes confirms that speedup ratios vary significantly with  $n$  for all three engines ( $p < 0.001$ ), indicating that dataset size is a meaningful moderator

| Engine                                                                                                   | $n$     | Speedup [95% CI]  | $p$ -value | Faster |
|----------------------------------------------------------------------------------------------------------|---------|-------------------|------------|--------|
| <i>Classification</i>                                                                                    |         |                   |            |        |
| <b>CatBoost</b> <sup>†</sup>                                                                             | 500     | 1.33 [1.27, 1.40] | 0.002**    | Rust   |
|                                                                                                          | 1,000   | 1.36 [1.32, 1.40] | 0.002**    | Rust   |
|                                                                                                          | 10,000  | 1.18 [1.06, 1.32] | 0.020*     | Rust   |
|                                                                                                          | 50,000  | 1.18 [1.03, 1.33] | 0.049*     | Rust   |
|                                                                                                          | 100,000 | 1.08 [1.01, 1.15] | 0.049*     | Rust   |
| <b>XGBoost</b>                                                                                           | 500     | 1.11 [1.01, 1.21] | 0.106      | —      |
|                                                                                                          | 10,000  | 0.66 [0.57, 0.77] | 0.004**    | C++    |
|                                                                                                          | 100,000 | 0.50 [0.44, 0.56] | 0.002**    | C++    |
| <i>Overall (36 comparisons across 3 engines <math>\times</math> 6 sizes <math>\times</math> 2 tasks)</i> |         |                   |            |        |
| Rust significantly faster                                                                                |         | 8 / 36 (22%)      |            |        |
| C++ significantly faster                                                                                 |         | 17 / 36 (47%)     |            |        |
| No significant difference                                                                                |         | 11 / 36 (31%)     |            |        |

Table 6: Statistical significance of performance differences (Wilcoxon signed-rank, paired, two-sided). Selected rows shown. \* $p < 0.05$ , \*\* $p < 0.01$ . Bootstrap 95% CI for the speedup ratio ( $> 1$  = Rust faster). **CatBoost** classification is significantly faster at 5 of 6 sizes.

| Dataset ( $n \times p$ )                  | Learner       | <b>smelt-ml</b> | <b>scikit-learn</b> | $\Delta$ |
|-------------------------------------------|---------------|-----------------|---------------------|----------|
| <i>Classification (accuracy)</i>          |               |                 |                     |          |
| Wine (178 $\times$ 13)                    | Decision Tree | 0.898           | 0.893               | +0.005   |
|                                           | Random Forest | 0.983           | 0.977               | +0.006   |
|                                           | Logistic Reg. | 0.989           | 0.955               | +0.034   |
| Breast Ca. (569 $\times$ 30)              | Decision Tree | 0.930           | 0.910               | +0.020   |
|                                           | Random Forest | 0.958           | 0.956               | +0.002   |
|                                           | Logistic Reg. | 0.981           | 0.953               | +0.028   |
| Digits (1,797 $\times$ 64)                | Decision Tree | 0.851           | 0.855               | −0.004   |
|                                           | Random Forest | 0.974           | 0.978               | −0.004   |
| <i>Regression (RMSE, lower is better)</i> |               |                 |                     |          |
| Cal. Housing (20,640 $\times$ 8)          | Decision Tree | 0.728           | 0.715               | +0.013   |
|                                           | Random Forest | 0.488           | 0.503               | −0.015   |
|                                           | Ridge         | 0.726           | 0.728               | −0.002   |

Table 7: 5-fold CV accuracy/RMSE on real datasets from  $n = 178$  to  $n = 20,640$ . Both frameworks use 5-fold CV with seed 42.

of relative performance.

This analysis was performed using **smelt-ml**’s built-in statistical testing module (`stats::wilcoxon_signed_rank`, `stats::bootstrap_ci`), demonstrating the framework’s integrated approach to model comparison.

#### 4.5. Accuracy validation

Table 7 validates correctness against **scikit-learn** on standard datasets.

All comparisons are within  $\pm 3.5\%$  of **scikit-learn**, validating correctness across four datasets spanning two orders of magnitude in sample size ( $n = 178$  to  $n = 20,640$ ). The Logistic Regression advantage on Wine and Breast Cancer is due to automatic feature standardization

in **smelt-ml**.

## 5. Integrated spatial ML pipeline

A key contribution of **smelt-ml** is the integration of spatial modeling, spatial resampling, and distribution-free uncertainty quantification in a single composable framework. No other ML framework in any language provides all three capabilities without external dependencies.

### 5.1. Components

**Geographical-XGBoost** (Grekousis 2025) extends gradient boosting with spatial awareness: bi-square kernel weights define local neighborhoods, per-location models capture spatially non-stationary relationships, and an adaptive  $\alpha$  weight blends local and global predictions. This is the first implementation outside the original Python package (Grekousis 2024).

**Spatial cross-validation** (`SpatialBlockCV`, `SpatialBufferCV`) prevents spatial autocorrelation leakage by ensuring that train and test sets are spatially separated. Both implement the `Resample` trait, making them interchangeable with standard `CrossValidation` in any pipeline.

**Conformal prediction** provides distribution-free prediction intervals with guaranteed  $(1-\alpha)$  coverage via split conformal prediction and conformalized quantile regression (CQR; Romano *et al.* 2019). The `ConformalRegressor` wraps any `TrainedModel` without retraining.

### 5.2. Why integration matters

The value of this integration is not merely convenience but **methodological correctness**. In practice, spatial ML workflows are error-prone because spatial leakage, a well-documented source of optimism bias in geospatial prediction (Roberts, Bahn, Ciuti, Boyce, Elith, Guillerá-Arroita, Hauenstein, Lahoz-Monfort, Schröder, Thuiller, Warton, Wintle, Hartig, and Dormann 2017), requires that the resampling strategy be spatially aware — yet most ML frameworks treat resampling and modeling as independent concerns. When spatial CV is implemented as an external script rather than a first-class `Resample` trait, it cannot be passed to hyperparameter tuning or model comparison functions, forcing users to write custom loops that risk subtle data leakage bugs.

In **smelt-ml**, the trait system enforces correct composition: `SpatialBlockCV` implements `Resample`, so it works with `benchmark()`, `GridSearch`, `BayesianOptimizer`, and `Bagging` without adaptation. Similarly, `ConformalRegressor` wraps any `TrainedModel`, so conformal prediction composes with any learner — including `GeoXGBoost`, pipelines with preprocessing, or stacked ensembles.

Table 8 quantifies the integration gap with Python.

### 5.3. Case study: housing price prediction in King County, WA

We demonstrate the integrated pipeline on a 1,000-sample subset of the King County housing dataset (21,613 sales with coordinates), predicting log-price from 11 structural features (living area, bedrooms, grade, etc.) *without* including latitude/longitude as model features. This

| Capability              | <b>smelt-ml</b>      | Python equivalent                 |
|-------------------------|----------------------|-----------------------------------|
| Spatial regression      | GeoXGBoost           | <b>geoxgboost</b>                 |
| Spatial CV              | SpatialBlockCV       | custom code or <b>spacv</b>       |
| Conformal prediction    | ConformalRegressor   | <b>MAPIE</b> or <b>crepes</b>     |
| Gradient boosting       | XGBoost              | <b>xgboost</b>                    |
| Preprocessing pipeline  | Pipeline             | <b>scikit-learn</b>               |
| Statistical validation  | stats module         | <b>scipy.stats</b>                |
| Packages required       | 1                    | 5–6                               |
| API consistency         | Trait-based, uniform | Mixed APIs                        |
| Data format conversion  | None                 | DataFrame $\leftrightarrow$ array |
| Thread-safety guarantee | Compile-time         | Runtime (GIL)                     |

Table 8: Integration comparison for a spatial ML workflow with uncertainty quantification.

isolates the contribution of spatial modeling from simple coordinate memorization.

```

let task = CsvLoader::from_path("data/king_county_1k.csv")
    .target("log_price").load_regress()?;
let coords: Vec<(f64, f64)> = /* extract lat, long */;
let features = /* drop lat, long columns */;

// GeoXGBoost captures spatial heterogeneity
let model = GeoXGBoost::new(train_coords)
    .with_bandwidth(50)
    .train_regress(&task)?;

// Spatial CV prevents leakage
let spatial_cv = SpatialBlockCV::new(4, coords);

// Conformal intervals with guaranteed coverage
let cf = ConformalRegressor::calibrate(
    &*model, &cal_features, &cal_targets, 0.1)?;

```

Table 9 shows three key findings.

**Spatial heterogeneity.** Adding coordinates as features improves **XGBoost** RMSE by 32%, confirming that housing price drivers vary strongly by location — precisely the scenario where Geographical-**XGBoost**’s local models provide value.

**Spatial leakage.** Standard 5-fold CV yields RMSE = 0.324, while **SpatialBlockCV** yields RMSE = 0.424 — a **31% optimism bias** from spatial autocorrelation leakage. This magnitude is consistent with [Roberts \*et al.\* \(2017\)](#), who report 10–40% bias in geospatial applications. Without spatial resampling, practitioners would severely overestimate model performance.

**Conformal coverage.** With  $\alpha = 0.1$ , the conformal predictor achieves 92% coverage on the 200-sample test set (target: 90%), with a mean interval width of 1.13 log-dollars. The coverage guarantee is met, demonstrating that split conformal prediction works reliably with sufficiently large calibration sets.

| Finding                                                  | Value        | Interpretation                 |
|----------------------------------------------------------|--------------|--------------------------------|
| <i>Model comparison (holdout, no coords in features)</i> |              |                                |
| XGBoost RMSE                                             | 0.337        | Baseline (no spatial info)     |
| XGBoost + coords RMSE                                    | 0.231        | 32% improvement with location  |
| <i>Spatial validation</i>                                |              |                                |
| Random CV RMSE                                           | 0.324        | Optimistic (spatial leakage)   |
| Spatial Block CV RMSE                                    | 0.424        | Honest (no leakage)            |
| Leakage bias                                             | 31%          | Random CV underestimates error |
| <i>Uncertainty quantification</i>                        |              |                                |
| Conformal coverage                                       | 92%          | Target: 90% — guarantee met    |
| Interval width                                           | 1.13 log(\$) | Calibrated uncertainty         |

Table 9: Case study results on King County housing (1,000 samples, 11 features, log-price target).

#### 5.4. Additional unique algorithms

**Causal Forest.** Honest splitting following [Athey and Imbens \(2016\)](#); [Wager and Athey \(2018\)](#). Estimates conditional average treatment effects (CATE)  $\tau(x) = E[Y(1) - Y(0) | X = x]$  with 95% confidence intervals. First implementation in a Rust ML framework.

**Oblique Forest (SPORF).** Sparse random projections following [Tomita et al. \(2020\)](#), enabling oblique decision boundaries.

**Streaming.** Hoeffding Trees for online learning on data streams, using the Hoeffding bound for statistically sound split decisions.

## 6. Summary and discussion

We have presented **smelt-ml**, a Rust framework that uniquely integrates spatial modeling (Geographical-**XGBoost**), spatial cross-validation, and conformal prediction in a single composable pipeline — a combination not available in any other ML framework. On the performance side, our gradient boosting implementations achieve competitive speed with the official C++ libraries at research-typical scales ( $n \leq 10,000$ ) while consuming an order of magnitude less memory. The performance advantage at small scales stems from Rust’s ownership model, which guarantees no aliasing (&mut is unique) and enables LLVM to apply more aggressive auto-vectorization than is possible with C++, where pointer aliasing must be assumed unless explicitly annotated. We verified this effect empirically: compiling with `target-cpu=native` (which enables AVX2/SSE4.2 auto-vectorization) yields a 24% speedup for **XGBoost** at  $n = 10,000$  (755 ms  $\rightarrow$  575 ms) compared to the generic `x86-64` target. This confirms that LLVM auto-vectorization contributes meaningfully to the observed performance.

**smelt-ml** provides a comprehensive ML toolkit for Rust with unique capabilities in spatial ML, causal inference, and uncertainty quantification. The framework contains zero `unsafe` blocks across 17,000+ lines of code, demonstrating that memory safety and high performance are not mutually exclusive. The test suite comprises 324 tests (plus 8 ignored benchmarks requiring external datasets) covering all learners, preprocessing, resampling, tuning, serializa-

tion, and edge cases (empty datasets, single-sample inputs, NaN features, multiclass targets, convergence checks against **scikit-learn**).

Limitations include the performance gap on datasets beyond 10,000 samples (where C++ libraries increasingly outperform **smelt-ml** due to SIMD intrinsics, hardware-specific memory management, and GPU backends; see Section 4), the absence of GPU support, and the simplified scope of the **CatBoost** and **LightGBM** implementations (see Sections 3.4–3.5).

Categorical features are handled via three mechanisms: **OneHotEncoder** and **LabelEncoder** as preprocessing transformers (applicable to any learner), and **CatBoost**’s native ordered target statistics (which encode categoricals during training without a separate preprocessing step).

Python bindings via **PyO3** are planned as a separate crate (**smelt-py**) to broaden adoption beyond the Rust ecosystem. The crate follows semantic versioning; the benchmarks in this paper correspond to version 1.3.0.

The software is available at <https://crates.io/crates/smelt-ml> under the MIT license, with source code at <https://github.com/franciscoparrrao/smelt>.

## Computational details

The results in this paper were obtained using Rust 1.89 with the **ndarray** 0.16, **rayon** 1.10, and **serde** 1.0 crates. Official C++ libraries were accessed via Python 3.12 with **xgboost** 3.1, **lightgbm** 4.6, and **catboost** 1.2. **scikit-learn** 1.8 was used for accuracy validation. Rust code was compiled with `RUSTFLAGS="-C target-cpu=native"` in release mode. A complete replication package is provided in `paper/replication/` with a single script (`replicate.sh`) that reproduces all benchmark tables and figures.

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